

Enterprise IT Lab

Project Documentation

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github.com/Ayman6594/AD-Project

This document covers:

- **Active Directory:** Domain Controller, DNS, DHCP, GPOs
- **Client-01:** Windows 11 domain join and GPO verification
- **Monitor-01:** Prometheus, Node Exporter, Grafana live dashboard

Lab Architecture

A fully functional enterprise IT environment running on VMware Workstation. Three virtual machines communicate over a Host-Only internal network with internet access via NAT. The lab replicates real-world corporate infrastructure found in companies across Morocco and internationally.

Virtual Machines and Network

VM	OS	IP Address	RAM	Role
DC01	Windows Server 2022	192.168.1.10	4 GB	Domain Controller, DNS, DHCP
Client-01	Windows 11	192.168.1.x (DHCP)	4 GB	Domain workstation
Monitor-01	Ubuntu 24.04 LTS	192.168.1.101	4 GB	Prometheus, Grafana, Wazuh

Services and Access URLs

Service	URL / Address	Credentials
Grafana	http://192.168.1.101:3000	admin / User@123456
Prometheus	http://192.168.1.101:9090	No auth
Node Exporter	http://192.168.1.101:9100/metrics	No auth
DC01 RDP	192.168.1.10	IB\Administrator
Domain Login	IB\username	User@123456

1. Active Directory

Windows Server 2022 was installed and promoted as the Domain Controller for the IB.local forest. The entire configuration was automated using a single PowerShell script — from OU structure to GPO deployment — demonstrating infrastructure-as-code in a Windows environment.

Domain Information

- Domain name: IB.local
- NetBIOS name: IB
- Forest and domain functional level: Windows Server 2016
- Domain Controller: DC01 — 192.168.1.10
- DNS: integrated with AD, pointing to 127.0.0.1

Automation Script

A PowerShell script (AD-Setup.ps1) was written to deploy the full AD configuration automatically. It runs in under 5 minutes and creates the complete environment from scratch.

```
Set-ExecutionPolicy Unrestricted -Force  
.\AD-Setup.ps1
```

What the script creates:

- 6 Organizational Units: IB-Company, IT, Finance, Management, Users, Computers, Service-Accounts
- 4 Security Groups: GRP-IT, GRP-Finance, GRP-Management, GRP-AllUsers
- 5 Domain Users assigned to their respective OUs and groups
- DHCP Server installed, authorized, and scope configured
- 6 GPOs created and linked to their target OUs

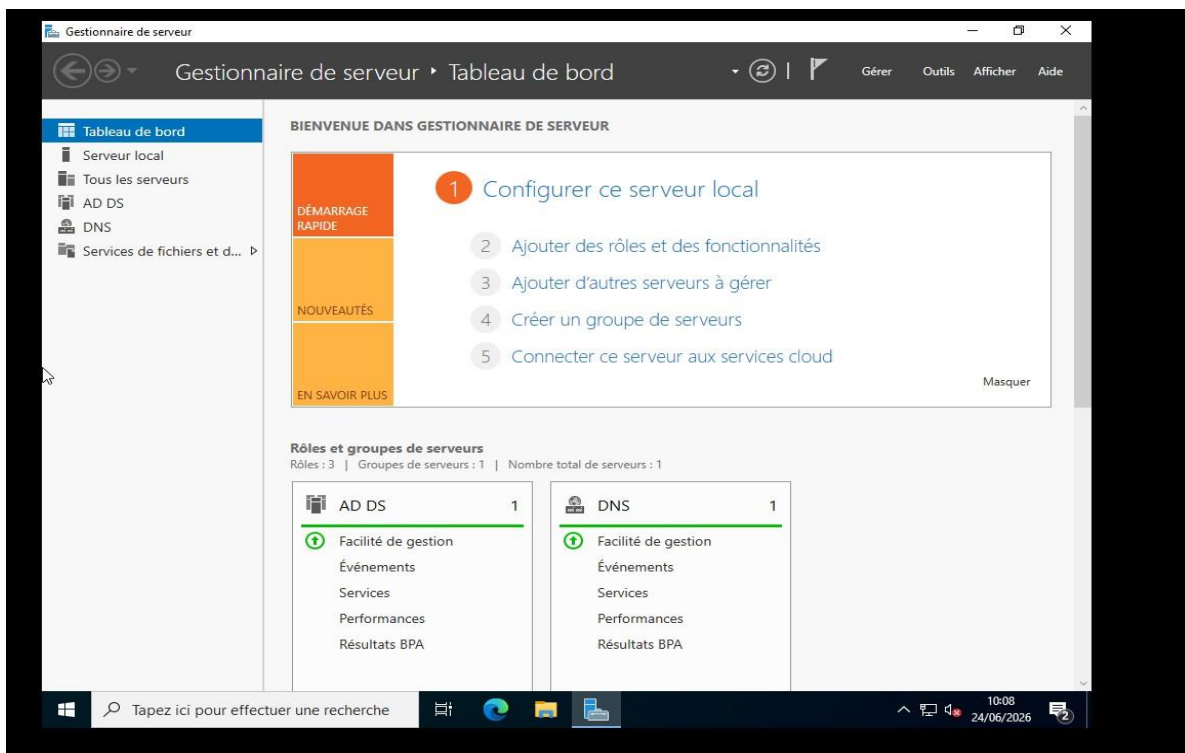


Figure 1 — Server Manager showing AD DS and DNS services active and green

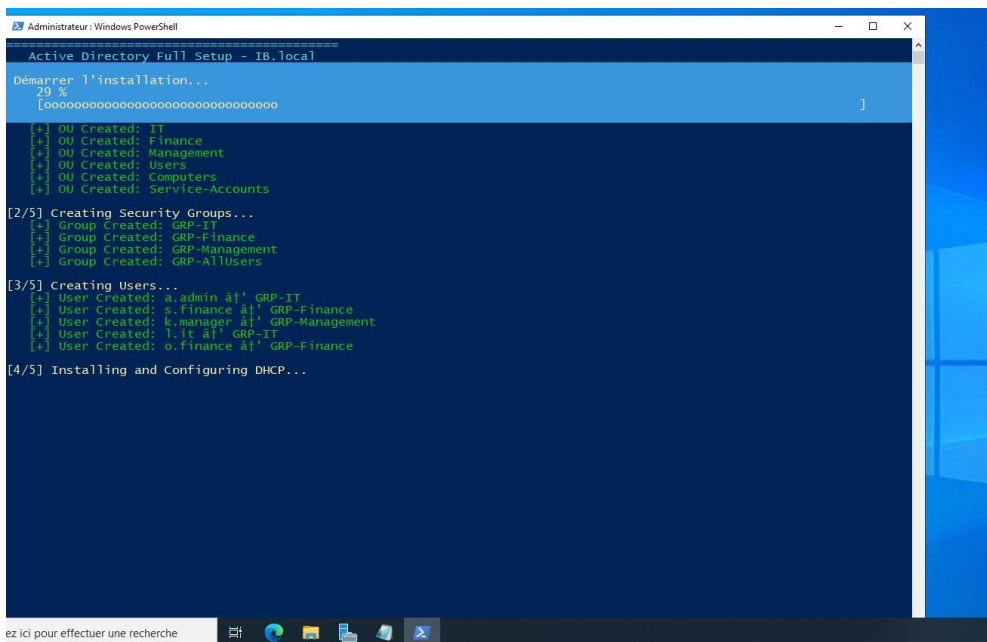


Figure 2 — PowerShell script running: OUs, Groups and Users created automatically

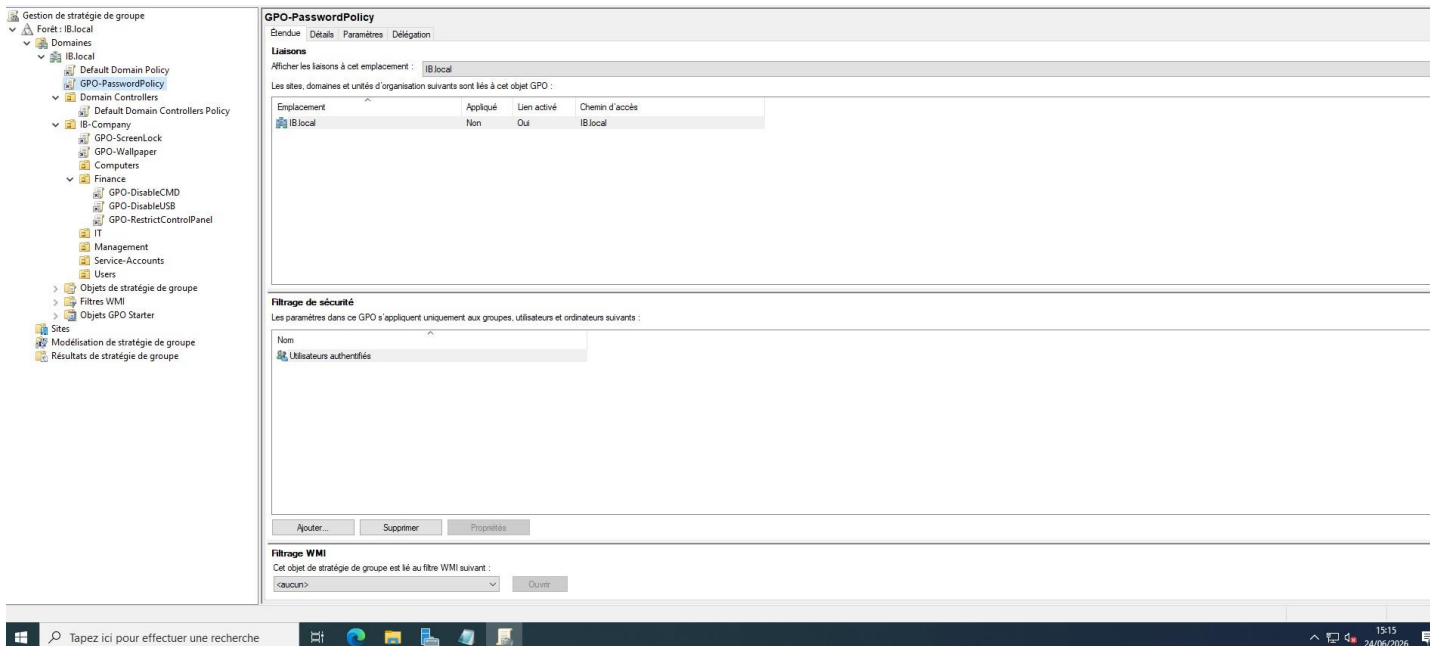


Figure 3 — Group Policy Management Console showing all 6 GPOs linked to their OUs

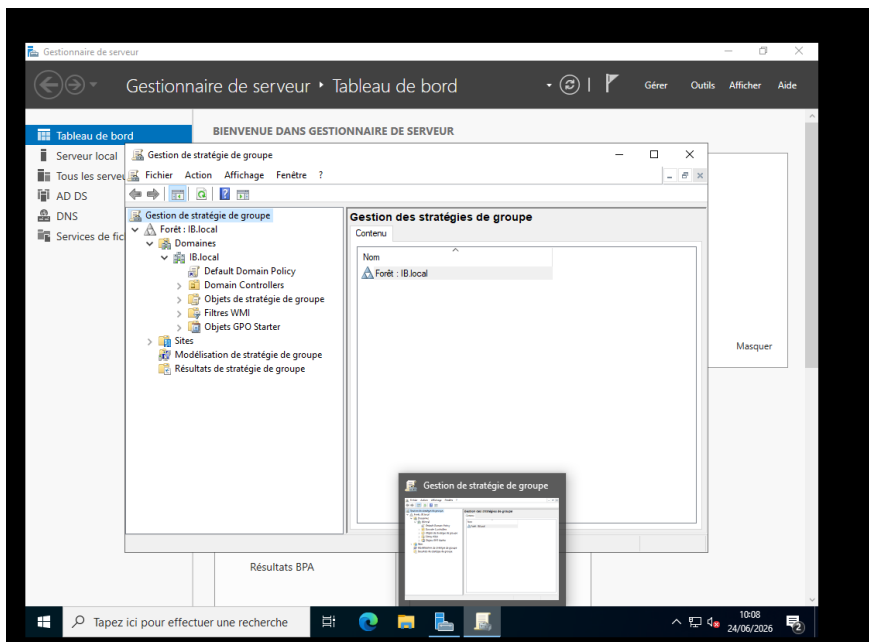


Figure 4 — Full OU and GPO tree structure in IB.local domain

Users Created

- a.admin (Ayman Admin) — IT — GRP-IT — IT Administrator
- l.it (Leila IT) — IT — GRP-IT — IT Support
- s.finance (Sara Finance) — Finance — GRP-Finance — Financial Analyst
- o.finance (Omar Finance) — Finance — GRP-Finance — Accountant
- k.manager (Karim Manager) — Management — GRP-Management — Department Manager

GPOs Deployed

- GPO-PasswordPolicy → IB.local: strong password + 90-day expiry
- GPO-ScreenLock → IB-Company: screen locks after 10 minutes
- GPO-Wallpaper → IB-Company: company wallpaper on all machines
- GPO-DisableUSB → Finance: USB storage blocked
- GPO-RestrictControlPanel → Finance: Control Panel inaccessible
- GPO-DisableCMD → Finance: CMD and PowerShell blocked

GitHub

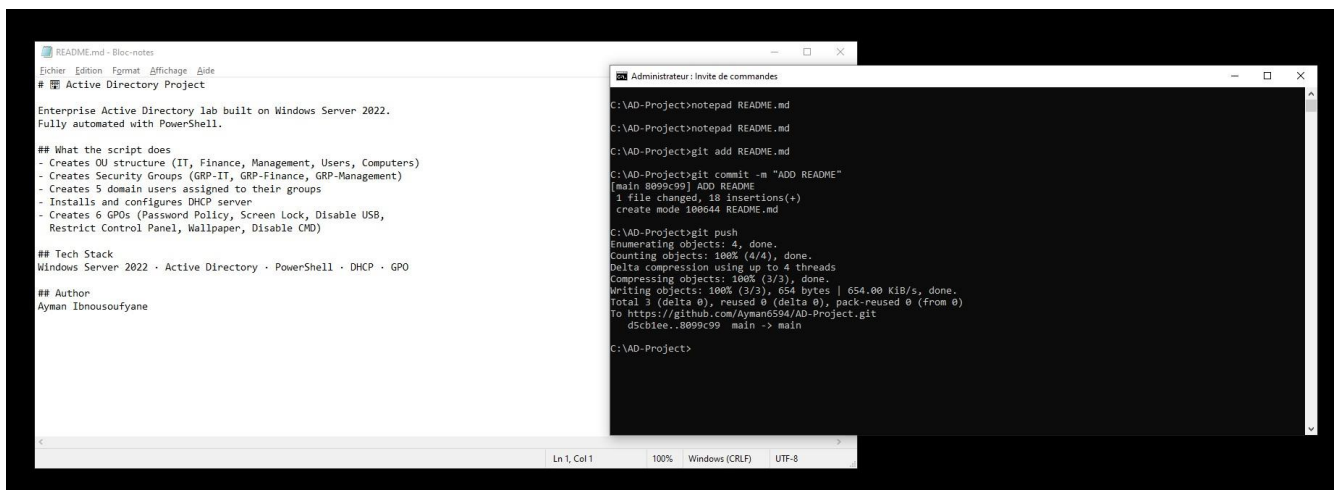


Figure 5 — GitHub repository AD-Project with PowerShell script and README

2. Client-01 — Domain Workstation

A Windows 11 virtual machine was joined to the IB.local domain. Domain users can log in from the workstation and all GPOs apply automatically at login — exactly as in a real corporate environment. The machine was given a Host-Only network adapter so it can reach DC01 at 192.168.1.10.

Domain Join Steps

- Added Host-Only network adapter in VMware settings
- Set DNS to 192.168.1.10 (DC01) on the Host-Only adapter
- System Properties → Change → Domain: IB.local
- Authenticated with IB\Administrator — restart
- Logged in with IB*.admin and IB\s.finance — confirmed working

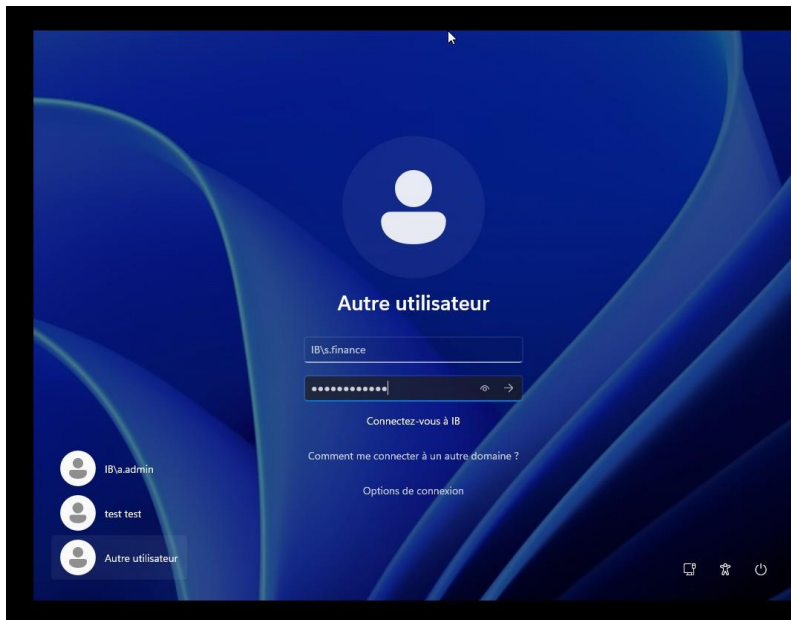


Figure 6 — Windows 11 login screen: IB domain users visible — IB*.admin and IB\s.finance

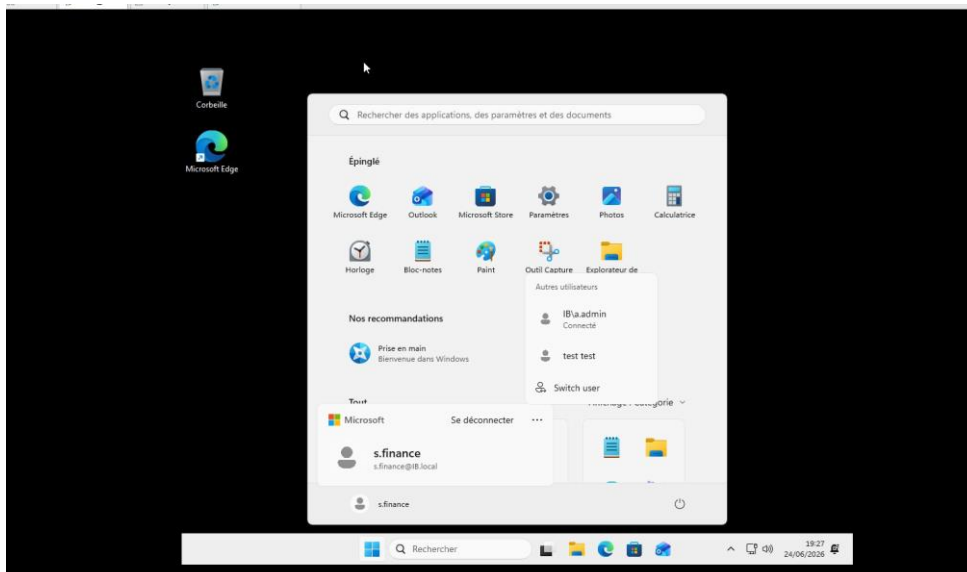


Figure 7 — Start menu logged in as s.finance@IB.local — domain account confirmed

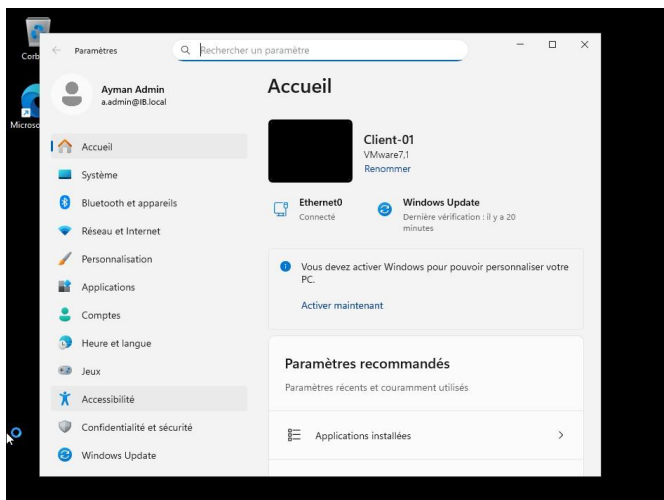


Figure 8 — System Settings showing a.admin@IB.local on Client-01 (machine: Client-01)

GPO Verification

After logging in as s.finance (Finance group), all Finance-targeted GPOs were tested and confirmed working on Client-01.

- CMD opened → immediately shows "disabled by your administrator" ✓
- Control Panel → "Operation cancelled due to restrictions on this computer" ✓
- USB storage device → blocked by the system ✓

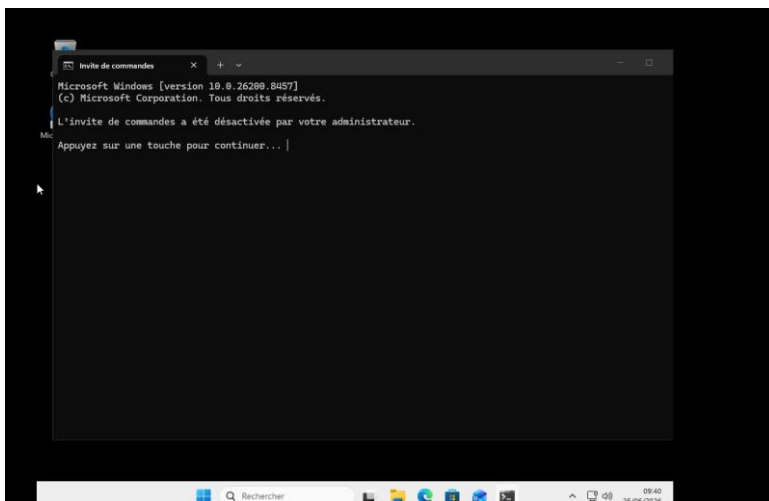


Figure 9 — CMD blocked: 'The command prompt has been disabled by your administrator'

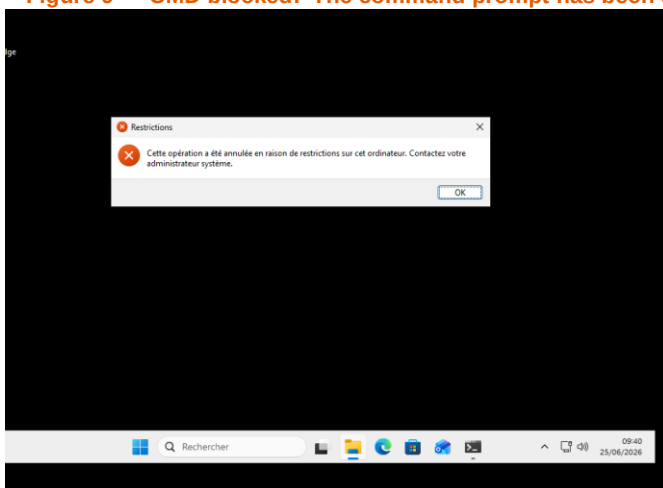


Figure 10 — Control Panel blocked: 'Operation cancelled due to restrictions'

3. Monitoring — Prometheus + Grafana

An Ubuntu 24.04 LTS server (Monitor-01) was deployed as a dedicated monitoring node. Using a single bash script, the full monitoring stack was installed and configured: Prometheus collects metrics every 15 seconds, Node Exporter exposes Linux system stats, and Grafana displays everything in a live dashboard accessible from any machine in the lab.

Ubuntu Server Installation

- Ubuntu 24.04 LTS ISO downloaded from ubuntu.com/server
- New VM created in VMware: 4GB RAM, 40GB disk, NAT + Host-Only adapters
- Static IP set via netplan: 192.168.1.101 on ens34 (Host-Only)
- Ping to DC01 (192.168.1.10) confirmed — network working

Netplan configuration applied

```
sudo nano /etc/netplan/00-installer-config.yaml
sudo chmod 600 /etc/netplan/00-installer-config.yaml
sudo netplan apply
ping 192.168.1.10
```

Installation Script

A bash script (install-monitoring.sh) installs and configures everything automatically. It creates system users, downloads binaries, writes systemd service files, and starts all services.

```
chmod +x ~/install-monitoring.sh
sudo ~/install-monitoring.sh
```

- Prometheus installed at /usr/local/bin/prometheus — port 9090
- Node Exporter installed at /usr/local/bin/node_exporter — port 9100
- Grafana installed via official apt repository — port 3000
- Wazuh Agent installed via packages.wazuh.com — connected to 192.168.1.20
- Windows Exporter installed on DC01 via PowerShell — port 9182



```
monitor-01@monitor-01: ~$ ./install-monitoring.sh
libfused2 libusb2
Veuillez utiliser « sudo apt autoremove » pour les supprimer.
Les NOUVEAUX paquets suivants seront installés :
  wazuh-agent
0 mis à jour, 1 nouvellement installés, 0 à enlever et 1 non mis à jour.
Il est nécessaire de prendre 11,2 Mo dans les archives.
Après cette opération, 49,0 Mo d'espace disque supplémentaires seront utilisés.
Réception de :1 https://packages.wazuh.com/4.x/apt/stable/main amd64 wazuh-agent amd64 4.14.5-1 [11,2 Mo]
11,2 Mo réceptionnés en 7s (1 969 ko/s)
Préconfiguration des paquets...
Sélection du paquet wazuh-agent précédemment désélectionné.
(Lecture de la base de données... 104100 fichiers et répertoires déjà installés.)
Préparation du dépaquetage de :./wazuh-agent_4.14.5-1_amd64.deb ...
Dépaquetage de wazuh-agent (4.14.5-1) ...
Paramétrage de wazuh-agent (4.14.5-1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
Created symlink /etc/systemd/system/multi-user.target.wants/wazuh-agent.service → /usr/lib/systemd/system/wazuh-agent.service.
[+] Wazuh Agent installed

=====
INSTALLATION COMPLETE!
=====

Services running:
Prometheus : http://192.168.1.101:9090
Node Export : http://192.168.1.101:9100
Grafana : http://192.168.1.101:3000
Wazuh Agent : connected to 192.168.1.20

Grafana login:
User : admin
Password: admin

Next: Open Grafana on Client-01 browser!

monitor-01@monitor-01:~$
```

Figure 11 — Bash script output: all 4 components installed and services started

Grafana Configuration

- Opened at <http://192.168.1.101:3000> from Client-01 browser
- Default login: admin / admin — changed password on first login
- Data source added: Prometheus at <http://localhost:9090>
- Dashboard imported from Grafana.com: Node Exporter Dashboard (ID 11074)
- Instance selected: localhost:9100 — Job: node-exporter

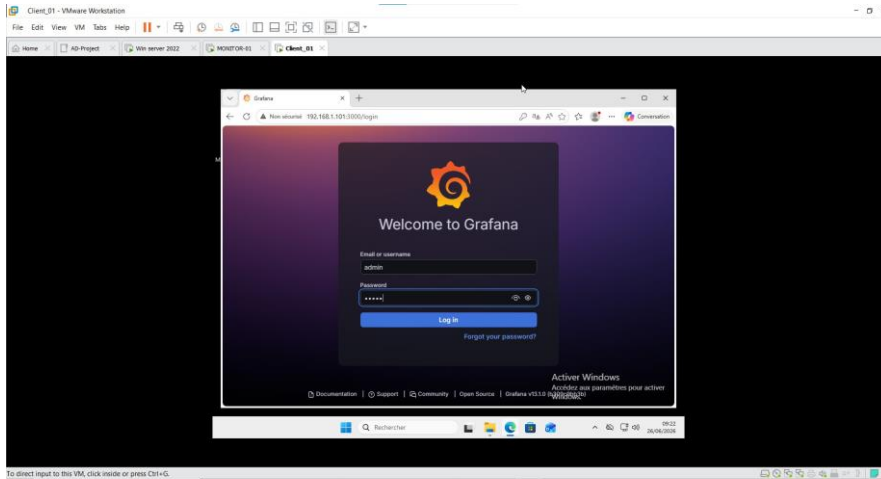


Figure 12 — Grafana login page accessed from Client-01 at <http://192.168.1.101:3000>

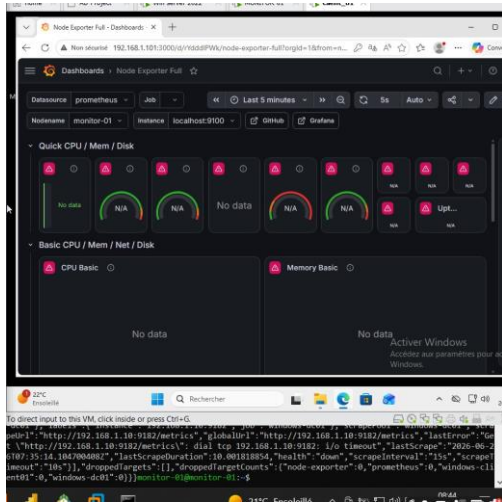


Figure 13 — Dashboard live: CPU 4.9%, RAM 30.9%, Disk 4.9GB used, instance monitor-01

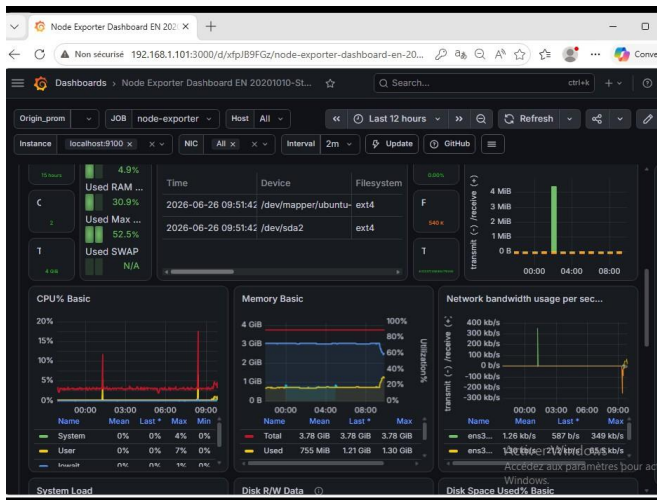


Figure 14 — Full dashboard: CPU Basic, Memory Basic, Network bandwidth, Disk R/W in real-time

Metrics collected every 15 seconds

- CPU usage per core — idle, user, system, iowait modes
- RAM total 3.78 GiB — used 1.21 GiB — cached
- Disk I/O read/write — /dev/mapper/ubuntu ext4 and /dev/sda2
- Network bandwidth — ens33 (NAT) and ens34 (Host-Only)
- System load average 1m / 5m / 15m
- Filesystem usage percentage

```
lines 1-20/20 (END)
monitor-01@monitor-01:~$ curl http://localhost:9090/api/v1/targets
curl: (6) Could not resolve host: localhost
monitor-01@monitor-01:~$ curl http://localhost:9090/api/v1/targets
{"status": "success", "data": {"activeTargets": [{"discoveredLabels": {"_address_": "localhost:9100", "_metrics_path_": "/metrics", "_scheme_": "http", "_scrape_interval_": "15s", "_scrape_timeout_": "10s", "job": "node-exporter"}, "labels": {"instance": "localhost:9100", "job": "node-exporter"}, "scrapePool": "node-exporter", "scrapeUrl": "http://localhost:9100/metrics", "globalUrl": "http://monitor-01:9100/metrics", "lastError": "", "lastScrape": "2026-06-26T07:35:31.988943739Z", "lastScrapeDuration": "0.018219046", "health": "up", "scrapeInterval": "15s", "scrapeTimeout": "10s"}, {"discoveredLabels": {"_address_": "localhost:9090", "_metrics_path_": "/metrics", "_scheme_": "http", "_scrape_interval_": "15s", "_scrape_timeout_": "10s", "job": "prometheus"}, "labels": {"instance": "localhost:9090", "job": "prometheus"}, "scrapePool": "prometheus", "scrapeUrl": "http://localhost:9090/metrics", "globalUrl": "http://monitor-01:9090/metrics", "lastError": "", "lastScrape": "2026-06-26T07:35:31.22868994Z", "lastScrapeDuration": "0.005261322", "health": "up", "scrapeInterval": "15s", "scrapeTimeout": "10s"}, {"discoveredLabels": {"_address_": "192.168.1.101:9182", "_metrics_path_": "/metrics", "_scheme_": "http", "_scrape_interval_": "15s", "_scrape_timeout_": "10s", "job": "windows-client01"}, "labels": {"instance": "192.168.1.101:9182", "job": "windows-client01"}, "scrapePool": "windows-client01", "scrapeUrl": "http://192.168.1.101:9182/metrics", "globalUrl": "http://192.168.1.101:9182/metrics", "lastError": "Get \"http://192.168.1.101:9182/metrics\": dial tcp 192.168.1.101:9182: connect: connection refused", "lastScrape": "2026-06-26T07:35:34.51827449Z", "lastScrapeDuration": "0.000657902", "health": "down", "scrapeInterval": "15s", "scrapeTimeout": "10s"}, {"discoveredLabels": {"_address_": "192.168.1.10:9182", "_metrics_path_": "/metrics", "_scheme_": "http", "_scrape_interval_": "15s", "_scrape_timeout_": "10s", "job": "windows-dc01"}, "labels": {"instance": "192.168.1.10:9182", "job": "windows-dc01"}, "scrapePool": "windows-dc01", "scrapeUrl": "http://192.168.1.10:9182/metrics", "globalUrl": "http://192.168.1.10:9182/metrics", "lastError": "Get \"http://192.168.1.10:9182/metrics\": dial tcp 192.168.1.10:9182: i/o timeout", "lastScrape": "2026-06-26T07:35:14.104700488Z", "lastScrapeDuration": "10.001818854", "health": "down", "scrapeInterval": "15s", "scrapeTimeout": "10s"}], "droppedTargetCounts": {"node-exporter": 0, "prometheus": 0, "windows-client01": 0, "windows-dc01": 0}}
monitor-01@monitor-01:~$
```

Figure 15 — Prometheus targets: node-exporter UP, windows exporters configured

What's Next

The lab continues to grow. Each upcoming step adds a new enterprise layer on top of what is already running.

Wazuh SIEM

- Wazuh Manager deployed on Monitor-01
- Agents on DC01 and Client-01 sending events
- Real-time alerts: failed logins, GPO violations, USB attempts
- Dashboard for security events and threat detection

Graylog — Centralized Logging

- Collects Windows Event Logs from DC01 and Client-01
- Collects Linux syslogs from Monitor-01
- Search, filter and correlate logs from all machines in one place

VPN + AD Integration

- Connect the SSL VPN (ocserv) to Active Directory via LDAP
- Domain users authenticate to VPN with their AD credentials
- Same principle as FortiClient with LDAP/AD integration

Enterprise IT Lab — Ibousoufyane Ayman

June 2026 | VMware Workstation | Windows Server 2022 · Windows 11 · Ubuntu 24.04

Stack: Active Directory · GPO · Prometheus · Grafana · Wazuh

Repository: github.com/Ayman6594/AD-Project